Cambridge Assessment International Education

Cambridge International AS & A Level

MATHEMATICS **9709/51**

Paper 5 Probability & Statistics 1 **May/June 2023**

MARK SCHEME

Maximum Mark: 50

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2023 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

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Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:
Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:
Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:
Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:
Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:
Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:
Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

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Mathematics Specific Marking Principles	
1	Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
2	Unless specified in the question, answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
3	Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
4	Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
5	Where a candidate has misread a number in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 mark for the misread.
6	Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

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Mark Scheme Notes

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

Types of mark

M Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.

A Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).

B Mark for a correct result or statement independent of method marks.

DM or **DB** When a part of a question has two or more 'method' steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly, when there are several B marks allocated. The notation DM or DB is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.

FT Implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only.

- A or B marks are given for correct work only (not for results obtained from incorrect working) unless follow through is allowed (see abbreviation FT above).
- For a numerical answer, allow the A or B mark if the answer is correct to 3 significant figures or would be correct to 3 significant figures if rounded (1 decimal place for angles in degrees).
- The total number of marks available for each question is shown at the bottom of the Marks column.
- Wrong or missing units in an answer should not result in loss of marks unless the guidance indicates otherwise.
- Square brackets [ ] around text or numbers show extra information not needed for the mark to be awarded.

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Abbreviations

AEF/OE	Any Equivalent Form (of answer is equally acceptable) / Or Equivalent
AG	Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)
CAO	Correct Answer Only (emphasising that no 'follow through' from a previous error is allowed)
CWO	Correct Working Only
ISW	Ignore Subsequent Working
SOI	Seen Or Implied
SC	Special Case (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance)
WWW	Without Wrong Working
AWRT	Answer Which Rounds To

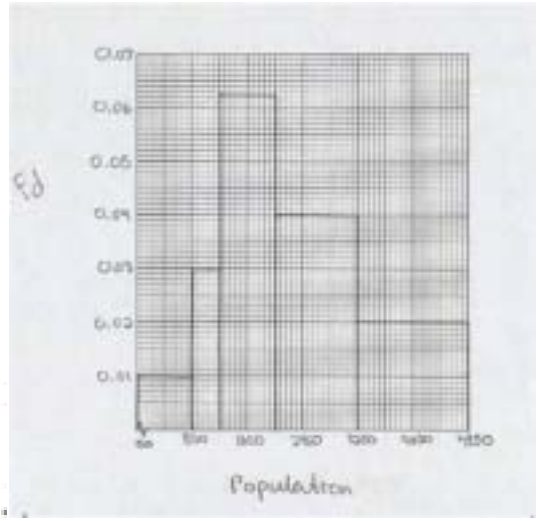
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Question	Answer	Marks	Guidance
1(a)	Var = $\left[\frac{\Sigma(x-q)^2}{50}-\left(\frac{\Sigma(x-q)}{50}\right)^2=\right]\frac{14235}{50}-\left(\frac{700}{50}\right)^2$ $[= 284.7 - 196 = 88.7]$	**M1**	$\frac{14235}{a}-\left(\frac{700}{a}\right)^2$; where a = 49, 50, 51.
	[sd = $\sqrt{88.7}$ =] 9.42	**A1**	9.4180677 rounded to at least 3SF.
		2	
1(b)	$\Sigma x - 50q = 700$ [2865 – 50q = 700]	**M1**	Forming equation with Σx, 50q and 700.
	$q = 43.3, 43\frac{3}{10}$	**A1**	If M0 scored, **SC B1** for 43.3 WWW.
		2	

Question	Answer	Marks	Guidance
2(a)	${}^{6}C_{3} \times {}^{8}C_{3}$	**M1**	${}^{6}C_{3} \times b$ or $c \times {}^{8}C_{3}$ seen. b, c integers $\geqslant 1$ (1 may be implied).
	1120	**A1**	
		2	
2(b)	**Method 1**		
	0 brothers $[{}^{3}C_{0}] \times {}^{11}C_{6}$ 462 1 brother ${}^{3}C_{1} \times {}^{11}C_{5}$ 1386 2 brothers ${}^{3}C_{2} \times {}^{11}C_{4}$ 990	**B1**	${}^{3}C_{x} \times {}^{11}C_{6-x}$, with $x = 1$ or 2 seen.
		M1	Add values of 3 correct scenarios, (may be identified by the appropriate calculations) no incorrect/repeated scenarios, condone use of permutations.
	2838	**A1**	Only dependent on the M mark. **SC B1** for the correct calculation or 2838 seen WWW.
	Method 2		
	${}^{14}C_{6} - {}^{11}C_{3}$ 3003 – 165	**B1**	${}^{14}C_{6} - d$, where d a positive integer.
		M1	$e - {}^{11}C_{3}$, where e is a positive integer >165.
	= 2838	**A1**	
		3	

Question	Answer	Marks	Guidance
3(a)	$\left[\frac{8!}{2!3!}=\right]$ 3360	**B1**	
		1	
3(b)	$\frac{6!}{2!2!}$	**M1**	$\frac{6!}{2!f!}$; f = 1, 2, 3.
	180	**A1**	
		2	
3(c)	$\left[P(OOO\|CC)=\frac{P(OOO\cap CC)}{P(CC)}=\right]$ $\frac{\frac{5!}{7!}}{3!}$	**M1**	$\frac{5!}{g}$ g a positive integer, g ≠ 3360, 1. Condone numerator of $\frac{5!}{3360g}$.
		M1	$\frac{\frac{h}{7!}}{3!}$ or $\frac{h}{\frac{8!}{3!}}$, where h is a positive integer. Condone division by 3360 in denominator.
	$=\frac{120}{840}, \frac{1}{7}, 0.143$	**A1**	0.1428571… to at least 3SF. If M0 scored **SC B1** for $\frac{1}{7}$ WWW.
		3	

Question	Answer	Marks	Guidance
4(a)	$\text{P}(Z > \frac{20-14.6}{5.2}) = P(Z > 1.03846)$	**M1**	Use of ± standardisation formula with 20, 14.6 and 5.2 not σ^2, not $\sqrt{\sigma}$, no continuity correction.
	1 – 0.8504	**M1**	Calculating the appropriate probability area (leading to their final answer).
	0.150	**A1**	0.1496, $0.149 < p \leqslant 0.15$[0] . Only dependent on the 2nd M mark so M0M1A1 possible. **SC B1** for $0.149 < p \leqslant 0.15$[0] if M0M0A0 awarded.
	[250 × *their* 0.1496 =] 37, 38	**B1 FT**	Strict FT *their* at least 4-figure probability seen anywhere (give BOD if they go on to use 0.150). Final answer must be positive integer, no approximation or rounding stated.
		4	
4(b)	$z_1 = \frac{14.5-\mu}{\sigma} = -0.842$ $z_2 = \frac{18.5-\mu}{\sigma} = -0.44$	**B1**	$-0.843 < z_1 < -0.841$ or $0.841 < z_1 < 0.843$.
		B1	$-0.441 < z_2 < -0.439$ or $0.439 < z_2 < 0.441$.
		M1	Use of the ±standardisation formula once with μ, σ and a z-value (not 0.20, 0.80, 0.67, 0.23, 0.5793, 0.7881, 0.7486, 0.591 or 1 - z i.e. 0.158 etc.). Condone continuity correction ±0.05, not $\sigma^2, \sqrt{\sigma}$.
	Solve, obtaining values for μ and σ. $\mu = 22.9, \quad \sigma = 9.95$	**M1**	Solve using the elimination method, substitution method or other appropriate approach to obtain values for both μ and σ.
		A1	AWRT 22.9, 9.95 .
		5	

Question	Answer	Marks	Guidance
5(a)	<table><tr><td>cw</td><td>800</td><td>400</td><td>800</td><td>1200</td><td>1600</td></tr><tr><td>fd</td><td>0.01</td><td>0.03</td><td>0.0625</td><td>0.04</td><td>0.02</td></tr></table> 	**M1**	At least 4 frequency densities calculated (F/cw, e.g. $\frac{8}{800}\left(\text{condone}\frac{8}{n}, 799 \leqslant n \leqslant 801\right)$) Accept unsimplified, may be read from graph using *their* scale.
		A1	All heights correct on graph.
		B1	Bar ends at 50, 850, 1250, 2050, 3250, 4850 read at the axis with a horizontal linear scale with at least 3 values indicated. $50 \leqslant$ horizontal scale $\leqslant 4850$.
		B1	Axes labelled frequency density (fd) and population (pop) OE, or in a title. Linear vertical scale, with at least 3 values indicated. Vertical axis must cover at least the range $0 \leqslant$ vertical axis $\leqslant 0.0625$. Axes may be reversed.
		4	
5(b)	2100 – 3200	**B1**	Accept 2050 – 3250 OE. Condone '4th interval'.
		1	
5(c)	3249 – 1250	**M1**	$2050 \leqslant$ UQ $\leqslant 3250 - 1250 \leqslant$ LQ $\leqslant 2050$.
	1999	**A1**	Condone 3250 – 1250 = 2000.
		2	

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th><th>Guidance</th></tr>
<tr><td rowspan="3">6(a)</td><td>[P($X = 3$) =] $\frac{3}{4}\times\left(\frac{1}{4}\right)^3\times 4$</td><td>M1</td><td>$\frac{3}{4}\times\left(\frac{1}{4}\right)^3\times q$; q a positive integer (1 may be implied).</td></tr>
<tr><td>$=\frac{3}{64}$</td><td>A1</td><td>AG.</td></tr>
<tr><td></td><td>2</td><td></td></tr>
<tr><td rowspan="3">6(b)</td><td rowspan="2"><table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>P($X = x$)</td><td>$\frac{81}{256}$</td><td>$\frac{27}{64}$</td><td>$\frac{27}{128}$</td><td>$\frac{3}{64}$</td><td>$\frac{1}{256}$</td></tr></table></td><td>B1</td><td>Either P(1) = $\frac{27}{64}$,0.421875
or P(2) = $\frac{27}{128}$,0.2109375 correct to at least 3SF.
Condone not in table.</td></tr>
<tr><td>B1 FT</td><td>Both values in table.
FT P(1) + P(2) = $\frac{81}{128}$,0.6328125 .</td></tr>
<tr><td></td><td>2</td><td></td></tr>
<tr><td rowspan="3">6(c)</td><td>[E(X) =] [$0\times\frac{81}{256}$] + $1\times$their $\frac{27}{64}+2\times$their $\frac{27}{128}+3\times\frac{12}{256}+4\times\frac{1}{256}$</td><td>M1</td><td>Correct method from their probability distribution table with at least 4 terms, $0 <$ their P(x) < 1, accept partially evaluated.
$= 0+\frac{27}{64}+\frac{54}{128}+\frac{36}{256}+\frac{4}{256}$</td></tr>
<tr><td>$= 1$</td><td>A1</td><td></td></tr>
<tr><td></td><td>2</td><td></td></tr>
</table>

Question	Answer	Marks	Guidance
6(d)	Mean = $96\times\frac{67}{256}=25.125$ Var = $96\times\frac{67}{256}\times\frac{189}{256}=18.549$	**B1**	25.125, $25\frac{1}{8}$ and 18.5493… to at least 3SF seen, allow unsimplified ($4.3068 \leqslant \sigma \leqslant 4.307$ implies correct variance).
	P($X < 20$) = P($Z < \frac{19.5-25.125}{\sqrt{18.549}}$)	**M1**	Substituting *their* μ and σ into ± standardisation formula (any number for 19.5). Condone σ^2 and $\sqrt{\sigma}$.
		M1	Using continuity correction 19.5 or 20.5 in *their* standardisation formula. Note: $\frac{\pm 5.625}{\sqrt{18.549}}$ seen gains M2 BOD.
	[= P($Z < -1.306$) = 1 – Φ (1.306)=] 1 – 0.9042 =	**M1**	Appropriate area Φ, from final process. Must be a probability.
	0.0958	**A1**	$0.0957 \leqslant p \leqslant 0.0958$. **SC B1** for $0.0957 \leqslant p \leqslant 0.0958$ if B1M0M0M1 scored.
		5	

Question	Answer	Marks	Guidance
7(a)	**Method 1**		
	[P($X < 6$) = P($X \leqslant 5$) =] $1 - 0.8^5$	**M1**	$1 - 0.8^r$, $r = 5, 6$.
	= 0.672	**A1**	
	Method 2		
	[P($X < 6$) = P($X = 1$) + P($X = 2$) + P($X = 3$) + P($X = 4$) + P($X = 5$) =] $\frac{1}{5} + \frac{4}{5} \times \frac{1}{5} + \left(\frac{4}{5}\right)^2 \times \frac{1}{5} + \left(\frac{4}{5}\right)^3 \times \frac{1}{5} + \left(\frac{4}{5}\right)^4 \times \frac{1}{5}$	**M1**	Condone an extra term $(\frac{4}{5})^5 \times \frac{1}{5}$. First, last and one of the 3 middle terms implies M1.
	= 0.672	**A1**	
		2	

Question	Answer	Marks	Guidance
7(b)	**Method 1**		
	$[1 - P(0, 1, 2)]$ $= 1 - ({}^{12}C_0\,(0.8)^{12} + {}^{12}C_1\,(0.2)(0.8)^{11} + {}^{12}C_2\,(0.2)^2\,(0.8)^{10})$ $[= 1 - (0.06872 + 0.20615 + 0.28347)]$	**M1**	One term ${}^{12}C_x\,(p)^x\,(1-p)^{12-x}$, $0 < p < 1,\ x \neq 0, 1, 2$.
		A1	Correct expression, accept unsimplified, no terms omitted, leading to final answer. Correct unsimplified expression or better.
	$= 0.442$	**B1**	$0.411 < p \leqslant 0.442$ WWW.
	Method 2		
	$[P(3,4,5,6,7,8,9,10,11,12) =]$ ${}^{12}C_3\,(0.2)^3\,(0.8)^9 + {}^{12}C_4\,(0.2)^4\,(0.8)^8 + \ldots + {}^{12}C_{11}\,(0.2)^{11}\,(0.8)^1 + {}^{12}C_{12}\,(0.2)^{12}$ $[= 0.23622 + 0.13288 + \ldots + 1.966 \times 10^{-7} + 4.096 \times 10^{-9}]$	**M1**	One term ${}^{12}C_x\,(p)^x\,(1-p)^{12-x}$, $0 < p < 1,\ x \neq 0, 1, 2$.
		A1	Correct expression, accept unsimplified, leading to final answer. Accept first, last and 8 of the middle terms.
	$=0.442$	**B1**	$0.411 < p \leqslant 0.442$.
		3	

Question	Answer	Marks	Guidance
7(c)	$(0.2)^5 \times 5!$	**M1**	$(0.2)^5 \times s$, s a positive integer. 1 may be implied.
		M1	$t \times 5!$ where $0 < t < 1$.
	$= 0.0384, \frac{24}{625}$	**A1**	
	Alternative Method for Question 7(c)		
	$\frac{{}^5C_1 \times {}^4C_1 \times {}^3C_1 \times {}^2C_1 \times [{}^1C_1]}{({}^5C_1)^5}$	**M1**	$({}^5C_1)^5$ or 5^5 as denominator.
		M1	${}^5C_1 \times {}^4C_1 \times {}^3C_1 \times {}^2C_1 \times [{}^1C_1]$ or 5! as numerator.
	$= 0.0384, \frac{24}{625}$	**A1**	
		3	